

Python Code

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1 print("Assignment : 5 (:::-----String Basics -----:::~)")
2 print('1. Filter vowels and consonants from the string "How are you sir".')
3 def filter_vowels_consonants(text) :
4     vowels = ""
5     consonants = ""
6     for ch in text :
7         if ch.isalpha() :
8             if ch.lower() in "aeiou" :
9                 vowels += ch
10            else :
11                consonants += ch
12        print("Vowels : ", vowels)
13        print("Consonants : ", consonants)
14    def main() :
15        text = "How are you sir"
16        filter_vowels_consonants(text)
17    main()
18    print('2. Count vowels and consonants in the string "How are you sir".')
19    def count_vowels_consonants(text) :
20        vowels_count = 0
21        consonants_count = 0
22        for ch in text :
23            if ch.isalpha() :
24                if ch.lower() in "aeiou" :
25                    vowels_count += 1
26            else :
27                consonants_count += 1
28        print("Vowels Counts : ", vowels_count)
29        print("Consonants Counts : ", consonants_count)
30    def main():
31        text = "How are you sir"
32        count_vowels_consonants(text)
33    main()
34    print('3. Reverse the string "How are you sir".')
35    def reverse_string(text) :
36        reverse_string = ""
37        for ch in text :
38            reverse_string = ch + reverse_string
39        return reverse_string
40    def main():
41        text = "How are you sir"
42        result = reverse_string(text)
43        print("Original String : " , text)
44        print("Reverse String :", result)
45    main()
46    print('4. Convert lowercase letters to uppercase in the string "How are you sir".')
47    def convert_to_upper(text) :
48        upper_text = ""
49        for ch in text :
50            if ch.islower() :
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51         upper_text += ch.upper()
52     else :
53         upper_text += ch
54     return upper_text
55 def main() :
56     sentence = "How are you sir"
57     result = convert_to_upper(sentence)
58     print("Original Sentence : ", sentence)
59     print("Converted into Upper Case : ", result)
60 main()
61 print('5. Remove duplicate letters from the string "this is python programming
place".')
62 def remove_duplicates(text) :
63     result = ""
64     for ch in text :
65         if ch not in result :
66             result += ch
67     return result
68 def main():
69     text = "this is python programming place"
70     output = remove_duplicates(text)
71     print("Original String :", text)
72     print("Removed Duplicates : ", output)
73 main()
74 print('6. Search for a specific character in the string "this is python programming
place".')
75 def search_character(text,char) :
76     found = False
77     for ch in text :
78         if ch == char :
79             found = True
80             break
81     if found :
82         print(char, "is present in the string")
83     else :
84         print(char, "is not present in the string")
85 def main():
86     data = "this is python programming place"
87     search_char = input("Enter the chracter to search : ")
88     search_character(data,search_char)
89 main()
90 print('7. Find the greatest and smallest characters from the string "venugopaliyer".')
91 def find_greatest_smallest(string) :
92     smallest = string[0]
93     greatest = string[0]
94     for ch in string :
95         if ch < smallest :
96             smallest = ch
97         if ch > greatest :
98             greatest = ch
99     print("Smallest Chracter in string : ", smallest)
100    print("Greatest Chracter in String : ", greatest)
101 def main():
102     text = "venugopaliyer"
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103     find_greatest_smallest(text)
104 main()
105 print('8. Count the total occurrences of a specific letter in the string\n"this is
python programming place".')
106 def count_occurrences(text,letter) :
107     count = 0
108     for ch in text :
109         if ch == letter :
110             count += 1
111     print(f"{letter} occurs {count} times")
112 def main():
113     string = "this is python programming place"
114     letter = input("Enter Letter to Search : ")
115     count_occurrences(string,letter)
116 main()
117 print('9. Replace "python" with "javascript" in the string "python developer python
engineer python holder".')
118 def replace_word(text) :
119     result = text.replace("python","javascript")
120     return result
121 def main():
122     data = "python developer python engineer python holder"
123     output = replace_word(data)
124     print("Original String : ")
125     print(data)
126     print("Updated String : ")
127     print(output)
128 main()
129 print('10. Print alternate letters from the string "How are you sir".')
130 def alternate_letter(text) :
131     result = ""
132     for i in range(len(text)) :
133         if i % 2 == 0 :
134             result += text[i]
135     return result
136 def main():
137     data = "How are you sir"
138     output = alternate_letter(data)
139     print("String with Alternate Letter : ", output)
140 main()
141 print('11. Convert the string "qwertyuiopasdfghjklzxcvbnm" to
"abcdefghijklmnopqrstuvwxyz".')
142 def convert_string(text):
143     result = ""
144     for ch in "abcdefghijklmnopqrstuvwxyz" :
145         for letter in text :
146             if ch == letter :
147                 result += letter
148     return result
149 def main():
150     data = "qwertyuiopasdfghjklzxcvbnm"
151     print("Original Data : ", data)
152     print("Converted String : ", convert_string(data))
153 main()
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154 print("Extra String Questions : ")
155 print('2. Check if the string is a palindrome (e.g., "madam" → Palindrome, "hello" →
Not palindrome)')
156 def check_palindrome(text) :
157     reverse = ""
158     for ch in text :
159         reverse = ch + reverse
160     if text == reverse :
161         print(text,"String is Palindrome")
162     else :
163         print(text,"String is not palindrome")
164 def main():
165     data = input("Enter the string to check : ")
166     check_palindrome(data)
167 main()
168 print('13. Count spaces, digits, alphabets, and special characters in "Python 3.9 is
awesome!!".')
169 def count_character(text) :
170     spaces = 0
171     digits = 0
172     alphabets = 0
173     special_character = 0
174     for ch in text :
175         if ch.isalpha():
176             alphabets += 1
177         elif ch.isdigit() :
178             digits += 1
179         elif ch.isspace():
180             spaces += 1
181         else :
182             special_character += 1
183     print("Alphabets : ",alphabets)
184     print("Specials Chracter : ", special_character)
185     print("Digits : ", digits)
186     print("Spaces : ", spaces)
187 def main() :
188     data = "Python 3.9 is awesome!!"
189     count_character(data)
190 main()
191 print('14. Find the longest word in the string "Python programming is interesting".')
192 def longest_word(text) :
193     words = text.split()
194     print(words)
195     longest = words[0]
196     for word in words :
197         if len(word) > len(longest) :
198             longest = word
199     return longest
200 def main() :
201     string = "Python programming is interesting"
202     result = longest_word(string)
203     print("Longest String : ", result)
204 main()
205 print('15. Capitalize the first letter of each word in "welcome to python world".')
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206 def capitalize_word(text) :
207     result = ""
208     for word in text.split() :
209         result += word.capitalize() + " "
210     return result
211 def main() :
212     string = "welcome to python world"
213     output = capitalize_word(string)
214     print("Original String : ", string)
215     print("Capitalize String : ", output)
216 main()
217 print('16. Remove all spaces from "How are you sir".')
218 def remove_spaces(text) :
219     result = ""
220     for ch in text :
221         if ch != " " :
222             result += ch
223     return result
224 def main() :
225     data = "How are you sir"
226     output = remove_spaces(data)
227     print("Original String : ", data)
228     print("After Removing Spaces in String : ", output)
229 main()
230 print('18. Sort characters alphabetically in "programming" → "aggimnopr".')
231 def sort_string(text):
232     text = list(text)
233     for i in range(len(text)):
234         for j in range(i + 1, len(text)):
235             if text[i] > text[j]:
236                 text[i], text[j] = text[j], text[i]
237     result = ""
238     for ch in text:
239         result += ch
240     print(result)
241 def main():
242     data = "programming"
243     sort_string(data)
244 main()
245 print('19. Swap cases of all letters in "Python Is Fun" → "pYTHON iS fUN".')
246 def swap_case(text):
247     result = ""
248     for ch in text:
249         if ch.islower():
250             result += ch.upper()
251         elif ch.isupper():
252             result += ch.lower()
253         else:
254             result += ch
255     print(result)
256 def main():
257     data = "Python Is Fun"
258     swap_case(data)
259 main()
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260 print("20. Find frequency of each character in \"banana\" → { 'b':1, 'a':3, 'n':2 }.")
261 def frequency(text):
262     checked = ""
263
264     for ch in text:
265         if ch not in checked:
266             count = 0
267
268             for letter in text:
269                 if ch == letter:
270                     count += 1
271
272             print(ch, ":", count)
273             checked += ch
274 def main():
275     data = "banana"
276     frequency(data)
277 main()
278 print('21. Remove vowels from "How are you sir" → "Hw r y sr".')
279 def remove_vowels(text):
280     result = ""
281     for ch in text:
282         if ch.lower() not in "aeiou":
283             result += ch
284     return result
285 def main():
286     data = "How are you sir"
287     output = remove_vowels(data)
288     print("Original String :", data)
289     print("After Removing Vowels :", output)
290 main()
291 print('22. Check if a substring exists in "Python programming" (e.g., "thon" → Found).')
292 def check_substring(text, sub):
293     if sub in text:
294         print("Found")
295     else:
296         print("Not Found")
297 def main():
298     data = "Python programming"
299     substring = input("Enter substring: ")
300     check_substring(data, substring)
301 main()
302 print('23. Print words in reverse order in "How are you sir" → "sir you are How".')
303 def reverse_words(text):
304     words = text.split()
305     for i in range(len(words) - 1, -1, -1):
306         print(words[i], end=" ")
307 def main():
308     data = "How are you sir"
309     reverse_words(data)
310 main()
311 print('24. Count words in the string "This is a python assignment".')
312 def count_words(text):

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313     count = 1
314     for ch in text:
315         if ch == " ":
316             count += 1
317     print("Total Words :", count)
318 def main():
319     data = "This is a python assignment"
320     count_words(data)
321 main()
322 print('25. Find the ASCII value of each character in "ABcd".')
323 def ascii_value(text):
324     for ch in text:
325         print(ch, ":", ord(ch))
326 def main():
327     data = "ABcd"
328     ascii_value(data)
329 main()
330 print('26. Convert a string into a list of words using "split()" (e.g., "Python is fun" → ["Python", "is", "fun"]).')
331 def convert_list(text):
332     words = text.split()
333     print(words)
334 def main():
335     data = "Python is fun"
336     convert_list(data)
337 main()
338 print('27. Join a list of words into a string using "join()" (e.g., ["Python", "is", "fun"] → "Python is fun").')
339 def join_string(words):
340     result = " ".join(words)
341     print(result)
342 def main():
343     data = ["Python", "is", "fun"]
344     join_string(data)
345 main()
346 print('28. Find the first non-repeating character in "swiss" → "w".')
347 def first_non_repeating(text):
348     for ch in text:
349         if text.count(ch) == 1:
350             print("First Non-Repeating Character :", ch)
351             break
352 def main():
353     data = "swiss"
354     first_non_repeating(data)
355 main()
356 print('29. Check if two strings are anagrams (e.g., "listen" and "silent" → Anagrams).')
357 def check_anagram(str1, str2):
358     if len(str1) != len(str2):
359         print("Not Anagrams")
360         return
361     for ch in str1:
362         if str1.count(ch) != str2.count(ch):
363             print("Not Anagrams")
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364         return
365     print("Anagrams")
366 def main():
367     string1 = input("Enter first string: ")
368     string2 = input("Enter second string: ")
369     check_anagram(string1, string2)
370 main()
371 print('30. Replace all spaces with hyphens (-) in "Python is easy to learn" → "Python-is-easy-to-learn".')
372 def replace_space(text):
373     result = ""
374     for ch in text:
375         if ch == " ":
376             result += "-"
377         else:
378             result += ch
379     return result
380 def main():
381     data = "Python is easy to learn"
382     output = replace_space(data)
383     print("Original String :", data)
384     print("Updated String :", output)
385 main()
386 print("Substring Related Questions")
387 print('31. Extract a substring from "Python Programming" → from index 0 to 6 should give "Python".')
388 def extract_substring(text):
389     result = ""
390     for i in range(0, 6):
391         result += text[i]
392     return result
393 def main():
394     data = "Python Programming"
395     output = extract_substring(data)
396     print("Original String :", data)
397     print("Substring :", output)
398 main()
399 print('32. Check if one string is a substring of another (e.g., "gram" is a substring of "Programming").')
400 def check_substring(text, sub):
401     found = False
402     for i in range(len(text) - len(sub) + 1):
403         if text[i:i + len(sub)] == sub:
404             found = True
405             break
406     if found:
407         print("Substring Found")
408     else:
409         print("Substring Not Found")
410 def main():
411     data = "Programming"
412     substring = input("Enter substring: ")
413     check_substring(data, substring)
414 main()
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415 print('33. Find all occurrences of a substring in "This is Python and Python is fun" →
Substring "Python".')
416 def find_occurrences(text, sub):
417     found = False
418     for i in range(len(text) - len(sub) + 1):
419         if text[i:i + len(sub)] == sub:
420             print("Found at index:", i)
421             found = True
422     if not found:
423         print("Substring Not Found")
424 def main():
425     data = "This is Python and Python is fun"
426     substring = input("Enter substring: ")
427     find_occurrences(data, substring)
428 main()
429 print('34. Replace a substring in "I like Python" → Replace "Python" with "Java".')
430 def replace_substring(text, old, new):
431     result = ""
432     for word in text.split():
433         if word == old:
434             result += new + " "
435         else:
436             result += word + " "
437     return result.strip()
438 def main():
439     data = "I like Python"
440     output = replace_substring(data, "Python", "Java")
441     print("Original String :", data)
442     print("Updated String :", output)
443 main()
444 print('35. Remove a substring from "HelloWorld" → Remove "World" → "Hello".')
445 def remove_substring(text):
446     result = ""
447     i = 0
448     while i < len(text):
449         if text[i:i+5] == "World":
450             i += 5
451         else:
452             result += text[i]
453             i += 1
454     print("Original String :", text)
455     print("Updated String :", result)
456 def main():
457     data = "HelloWorld"
458     remove_substring(data)
459 main()
460 print('36. Count occurrences of a substring in "banana" → Substring "ana" appears 2
times.')
461 def count_substring(text, sub):
462     count = 0
463     for i in range(len(text) - len(sub) + 1):
464         if text[i:i+len(sub)] == sub:
465             count += 1
466     print("Substring Count :", count)

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467 def main():
468     data = "banana"
469     substring = "ana"
470     count_substring(data, substring)
471 main()
472 print('37. Check if a string starts with a substring (e.g., "Python is easy" starts
with "Python").')
473 def check_startswith(text, sub):
474     found = True
475     if len(sub) > len(text):
476         found = False
477     else:
478         for i in range(len(sub)):
479             if text[i] != sub[i]:
480                 found = False
481                 break
482     if found:
483         print("String starts with", sub)
484     else:
485         print("String does not start with", sub)
486 def main():
487     data = "Python is easy"
488     substring = input("Enter substring: ")
489     check_startswith(data, substring)
490 main()
491 print('38. Check if a string ends with a substring (e.g., "Learn coding" ends with
"coding").')
492 def check_endswith(text, sub):
493     found = True
494     if len(sub) > len(text):
495         found = False
496     else:
497         start = len(text) - len(sub)
498         for i in range(len(sub)):
499             if text[start + i] != sub[i]:
500                 found = False
501                 break
502     if found:
503         print("String ends with", sub)
504     else:
505         print("String does not end with", sub)
506 def main():
507     data = "Learn coding"
508     substring = input("Enter substring: ")
509     check_endswith(data, substring)
510 main()
511 print('39. Split a string based on a substring (e.g., "apple,banana,grapes" → Split by
", " → ["apple", "banana", "grapes"]')
512 def split_string(text):
513     result = text.split(",")
514     print(result)
515 def main():
516     data = "apple,banana,grapes"
517     split_string(data)
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518 main()
519 print('40. Find the index of the first occurrence of a substring in "Programming is
    great" → Substring "is" → Index 12.')
520 def first_occurrence(text, sub):
521     found = False
522     for i in range(len(text) - len(sub) + 1):
523         if text[i:i+len(sub)] == sub:
524             print("First Occurrence Index :", i)
525             found = True
526             break
527     if not found:
528         print("Substring Not Found")
529 def main():
530     data = "Programming is great"
531     substring = input("Enter substring: ")
532     first_occurrence(data, substring)
533 main()
534 print('41. Find the index of the last occurrence of a substring in "Programming in
    Python Programming" → Substring "Programming".')
535 def last_occurrence(text, sub):
536     index = -1
537     for i in range(len(text) - len(sub) + 1):
538         if text[i:i+len(sub)] == sub:
539             index = i
540     if index != -1:
541         print("Last Occurrence Index :", index)
542     else:
543         print("Substring Not Found")
544 def main():
545     data = "Programming in Python Programming"
546     substring = input("Enter substring: ")
547     last_occurrence(data, substring)
548 main()
549 print('42. Extract substring after a specific word (e.g., "Welcome to Python World" →
    substring after "to" → "Python World").')
550 def substring_after_word(text, word):
551     found = False
552     result = ""
553     words = text.split()
554     for w in words:
555         if found:
556             result += w + " "
557         if w == word:
558             found = True
559     print("Original String :", text)
560     print("Substring :", result.strip())
561 def main():
562     data = "Welcome to Python World"
563     search_word = input("Enter the word: ")
564     substring_after_word(data, search_word)
565 main()
566 print('43. Extract substring before a specific word (e.g., "Welcome to Python World" →
    substring before "Python" → "Welcome to").')
567 def substring_before_word(text, word):

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568     result = ""
569     words = text.split()
570     for w in words:
571         if w == word:
572             break
573         result += w + " "
574     print("Original String :", text)
575     print("Substring :", result.strip())
576 def main():
577     data = "Welcome to Python World"
578     search_word = input("Enter the word: ")
579     substring_before_word(data, search_word)
580 main()
581 print('44. Check if two strings are rotations (cyclic substrings) of each other (e.g.,
"abcd" and "cdab" → Rotations).')
582 def check_rotation(str1, str2):
583     if len(str1) != len(str2):
584         print("Not Rotations")
585         return
586     temp = str1 + str1
587     found = False
588     for i in range(len(temp) - len(str2) + 1):
589         if temp[i:i + len(str2)] == str2:
590             found = True
591             break
592     if found:
593         print("Rotations")
594     else:
595         print("Not Rotations")
596 def main():
597     string1 = input("Enter First String : ")
598     string2 = input("Enter Second String : ")
599     check_rotation(string1, string2)
600 main()
601 print('45. Find the longest common substring between two strings (e.g., "abcdxyz" and
"xyzabcd" → Longest common substring = "abcd").')
602 def longest_common_substring(str1, str2):
603     longest = ""
604     for i in range(len(str1)):
605         temp = ""
606         for j in range(i, len(str1)):
607             temp += str1[j]
608             if temp in str2:
609                 if len(temp) > len(longest):
610                     longest = temp
611     print("Longest Common Substring :", longest)
612 def main():
613     string1 = input("Enter First String : ")
614     string2 = input("Enter Second String : ")
615     longest_common_substring(string1, string2)
616 main()

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